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United States Patent	12415246
Kind Code	B2
Date of Patent	September 16, 2025
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Protective cover assembled on abrasive belt hand tool

Abstract

A protective cover assembled on an abrasive belt hand tool. The protective cover comprises an annular base and a tightening component. The annular base comprises a notch, a first clamping arm positioned on one side of the notch, and a second clamping arm positioned on the other side. The first clamping arm is formed with a through hole, the second clamping arm is formed with an assembly hole, the assembly hole does not penetrating through the second clamping arm. The annular base comprises a first state for tightening a protective cover mounting portion when the first clamping arm and the second clamping arm narrow the notch, and the annular base comprises a second state that the first clamping arm and the second clamping arm do not narrow the notch. The tightening component comprises a pull rod penetrating through the through hole and a piece arranged on the pull rod.

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Appl. No.:	16/878258
Filed:	May 19, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210362288 A1	Nov. 25, 2021

Publication Classification

Int. Cl.: B24B21/18 (20060101); B24B23/06 (20060101)

U.S. Cl.:

CPC **B24B21/18** (20130101); **B24B23/06** (20130101);

Field of Classification Search

CPC: B24B (55/057); B24B (55/06); B24B (55/052); B24B (21/18); B24B (23/005); B24B (23/06); B24B (23/02); B24B (21/20); B24B (21/00); B24B (23/028); B24B (55/05); B24B (27/08)

USPC: 451/355; 451/451; 451/452

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to a protective cover assembled on an abrasive belt hand tool, in particular to a protective cover which enables an annular base to tighten a protective cover mounting portion under the action of a tightening component.

BACKGROUND OF THE INVENTION

(2) In the market, the structures of the abrasive belt machines mostly designed to be implemented with replaceable abrasive belts so that the abrasive belt can be replaced after being used for a long time or the abrasive belt machines can be matched with different abrasive bags according to the requirements, the related structures is referred to patents such as CN 110216557A, U.S. Ser. No. 10/618,139B and CN 208945846U.

(3) According to the drawings illustrated in patents CN 110216557A and U.S. Ser. No. 10/618,139B, it can be found that the conventional abrasive belt machine is provided with a quick-release component on the body of the abrasive belt machine to facilitate the removal of the abrasive belt. However, in the conventional structure, the abrasive belt is mostly arranged on an abrasive belt mounting seat, and the abrasive belt mounting seat is provided with two through holes respectively. The quick-release component penetrates through the two through holes and is provided with an operation piece on one side of the two through holes and a stop block on the other side of the two through holes. When the quick-release structure operated by the piece, the stop block pushes the abrasive belt mounting seat to move towards the direction facing the piece, so that the abrasive belt mounting seat and the body of the abrasive belt machine are tightened, and the abrasive belt is mounted on the body of the abrasive belt machine. When the abrasive belt mounting seat is to be disassembled, the piece is operated again, so that the stop block no longer limits the abrasive belt mounting seat, and the abrasive belt mounting seat is disengaged from the body of the abrasive belt machine. However, in the conventional quick-release structure, it is necessary to pay attention to the assembly directions of the parts during assembly, and thereby the abrasive belt mounting seat cannot be fixed quickly. In addition, according to the patent CN 208945846U, although the abrasive belt mounting seat is assembled through the thread structure, the abrasive belt mounting seat cannot be fixed quickly with this operation manner.

SUMMARY OF THE INVENTION

(4) The main purpose of the present invention is to solve the problem of the conventional structure that must be paid attention to the assembly directions of the parts during assembly and the mounting of the conventional structure is not convenient.

(5) In order to achieve the above object, the present invention provides a protective cover assembled on an abrasive belt hand tool, wherein the protective cover is assembled with a driving component of the abrasive belt hand tool and is provided for an abrasive belt to move therein. The driving component is provided with a driving shaft and a protective cover mounting portion formed at one end of the driving shaft. The protective cover comprises an annular base and a tightening component, and the annular base is arranged on the protective cover mounting portion and provided for the driving shaft to extend into the annular base. The annular base comprises a notch, a first clamping arm positioned on one side of the notch and a second clamping arm positioned on one side of the notch which is not provided with the first clamping arm and facing the first clamping arm. The first clamping arm is formed with a through hole, and the second clamping arm is formed with an assembly hole on a same extension line as the through hole. The assembly hole does not penetrate through the second clamping arm, and the annular base comprises a first state for tightening the protective cover mounting portion when the first clamping arm and the second clamping arm are getting closed, to narrow the notch and a second state for releasing the protective cover mounting portion when the first clamping arm and the second clamping arm do not narrow the notch. The tightening component comprises a pull rod penetrating through the through hole with one end fixed with the assembly hole and a piece arranged at one end of the pull rod, which is not fixed with the assembly hole. The piece moves with respect to a surface of the annular base by taking the pull rod as an axial center to switch the annular base in the first state or the second state.

(6) In one embodiment, the piece comprises a cam portion assembled with the pull rod and moving with respect to a surface of the first clamping arm when the piece is stressed and a shifting portion connected with the cam portion. An assembling position of the piece and the pull rod are deviated from a center of the cam portion.

- (7) In one embodiment, the pull rod comprises a coupling section for assembling the piece, a connecting section extending from the coupling section and penetrating through the through hole, and a threading section extending from the connecting section and fixed in the assembly hole. An inner wall of the assembly hole is formed with a thread for assembling the threading section.
- (8) In one embodiment, a length of the assembly hole is longer than a length of the through hole.
- (9) In one embodiment, the tightening component comprises a gasket sleeved on the pull rod and arranged between the piece and the first clamping arm.
- (10) In one embodiment, the protective cover comprises an arc-shaped receiving groove arranged on the first clamping arm and facing the piece.
- (11) In one embodiment, the annular base comprises an abrasive belt supporting seat extending from the second clamping arm towards a direction opposite to the annular base.
- (12) From the above mention, the invention has the following characteristics compared with prior art: the first clamping arm and the second clamping arm which are respectively positioned on two sides of the notch are formed on the annular base, and the assembly hole formed on the second clamping arm does not penetrate through the second clamping arm, so that the tightening component is fixed in the assembly hole. When implemented, the annular base is forced by the tightening component, so that the first clamping arm and the second clamping arm narrow the notch to tighten the protective cover mounting portion. The present invention provides a protective cover which is different from a conventional one in structure and is convenient for a user to mount the protective cover on the abrasive belt hand tool.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a structural schematic view of an embodiment of the present invention assembled with an abrasive belt hand tool.
- (2) FIG. 2 is a structural exploded schematic view of an embodiment of the present invention.
- (3) FIG. 3 is a structural cross-sectional schematic view in the first state of an embodiment of the present invention.
- (4) FIG. 4 is a structural cross-sectional schematic view in the second state of an embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (5) The detailed description and technical contents of the present invention are described with reference to the drawings as follows:
- (6) Referring to FIG. 1, the present invention provides a protective cover **10** assembled with an abrasive belt hand tool **100**. A basic structure of the abrasive belt hand tool **100** is described. The abrasive belt hand tool **100** is not limited to a pneumatic hand-held tool machine or an electric hand-held tool machine. The abrasive belt hand tool **100** includes a driving component **20** driven by gas or electricity, and an abrasive belt **30** driven by the driving component **20**. The driving component **20** is assembled with the protective cover **10**, and the driving component **20** includes a driving shaft **21** for driving the abrasive belt **30** and a protective cover mounting portion **22** formed at one end of the driving shaft **21**. The driving shaft **21** penetrates into the protective cover **10** and allows the abrasive belt **30** to be arranged in the protective cover **10**, and one side of the protective cover mounting portion **22** is used for assembling the protective cover **10**.
- (7) Referring to FIG. 1, FIG. 2, FIG. 3 and FIG. 4, the protective cover **10** comprises an annular base **11** and a tightening component **13**. Specifically, the annular base **11** is assembled with the protective cover mounting portion **22** and used for the driving shaft **21** to extend in the annular base **11**. The annular base **11** includes a notch **111**, a first clamping arm **112** positioned on one side of the notch **111**, and a second clamping arm **113**. The second clamping arm **113** is positioned on one side

of the notch **111** which is not provided with the first clamping arm **112** and faces the first clamping arm **112**. The first clamping arm **112** is formed with a through hole **114**, and the through hole **114** faces and communicates with the notch **111**. The second clamping arm **113** is formed with an assembly hole **115**, and the assembly hole **115** is positioned on a same extension line as the through hole **114** and also faces the notch **111**. The assembly hole **115** does not penetrate through the second clamping arm **113** and communicates with the notch **111** on one side. In one embodiment, a length of the assembly hole **115** is longer than the length of the through hole **114**. Further, the annular base **11** is assembled with the tightening component **13**, the annular base **11** is in a first state **116** when being forced by the tightening component **13**, and the annular base **11** is in a second state **117** without being forced upon by the tightening component **13**. In the first state **116**, the first clamping arm **112** and the second clamping arm **113** are getting closed to narrow the notch **111**; in a second state **117**, the first clamping arm **112** and the second clamping arm **113** do not narrow the notch **111**. On the other hand, the tightening component **13** comprises a pull rod **131** penetrating through the through hole **114** and a piece **132** connected with the pull rod **131**. The pull rod **131** passes through the notch **111** from the through hole **114** to enter the assembly hole **115**. One end of the pull rod **131** is fixed in the assembly hole **115**, and the other end of the pull rod **131** protrudes out of the through hole **114** to be exposed on the surface of the first clamping arm **112**. The piece **132** is arranged on one side of the first clamping arm **112** which does not face the second clamping arm **113**, and the piece **132** is connected with one end of the pull rod **131** which is not fixed in the assembly hole **115**. The piece **132** is movable with respect to the annular base **11** by taking the pull rod **131** as an axial center, so that the annular base **11** can switch to be in the first state **116** or the second state **117**.

(8) With reference to FIG. 3 and FIG. 4, the implementation of the protective cover **10** is described. First, it is assumed that the protective cover **10** is not fixed with the protective cover mounting portion **22** initially, that is, the annular base **11** does not tighten the protective cover mounting portion **22**. Please refer to FIG. 3. To fix the protective cover **10** with the protective cover mounting portion **22**, the piece **132** presses against the first clamping arm **112** to shift towards the direction of the second clamping arm **113**. At the same time, the pull rod **131** is also forced by the piece **132**, causing the second clamping arm **113** and the first clamping arm **112** get closed. At this point, the first clamping arm **112** and the second clamping arm **113** narrow the notch **111** so that the annular base **11** is in the first state **116**. The annular base **11** is acted upon by the tightening component **13** to tighten the protective cover mounting portion **22**, so that the protective cover **10** is fixed with the protective cover mounting portion **22**. Further, please refer to FIG. 4. When the piece **132** displaces with respect to the surface of the first clamping arm **112** by taking the pull rod **131** as the axial center, the piece **132** stop pressing against the first clamping arm **112**. At the same time, the pull rod **131** is also acted upon by the piece **132**, causing the second clamping arm **113** and the first clamping arm **112** stop getting closed, so that the first clamping arm **112** and the second clamping arm **113** back to original position. At this point, the first clamping arm **112** and the second clamping arm **113** stop pressing against the notch **111**, causing the annular base **11** to transfer to the second state **117** from the first state **116**. Thus, the annular base **11** stop pressing the protective cover mounting portion **22** and allows the protective cover **10** to be disengaged with respect to the protective cover mounting portion **22**. Further, if the protective cover **10** is to be fixed with the protective cover mounting portion **22** again, the annular base **11** can press the protective cover mounting portion **22** by merely operating the piece **132** again to transfer the annular base **11** to be in the first state **116** from the second state **117**.

(9) Accordingly, the annular base **11** is formed with the first clamping arm **112** and the second clamping arm **113** which are respectively positioned on two sides of the notch **111**, and the assembly hole **115** formed on the second clamping arm **113** does not penetrate through the second clamping arm **113**, so that the tightening component **13** is fixed in the assembly hole **115**. When the annular base **11** is acted upon by the tightening component **13**, causing the first clamping arm **112**

and the second clamping arm **113** to narrow the notch **111**, and thereby the annular base **11** pressing the protective cover mounting portion **22**. The present invention provides a protective cover **10** which is different from a conventional structure and is convenient for a user to mount the protective cover **10** on the abrasive belt hand tool **100**.

(10) In one embodiment, referring to FIG. 2, FIG. 3, and FIG. 4, the piece **132** includes a cam portion **133** assembled with the pull rod **131** and a shifting portion **134** connected with the cam portion **133**. The cam portion **133** contacts the surface of the first clamping arm **112**, and is forced by the piece **132**. The cam portion **133** displaces with respect to the surface of the first clamping arm **112** by taking the assembling position of the cam portion **133** and the pull rod **131** as an axial center. Further, the assembling position of the piece **132** and the pull rod **131** is deviated from a center of the cam portion **133**. As shown in FIG. 3, the assembling position of the piece **132** and the pull rod **131** is relatively higher than the center of the cam portion **133**. The piece **132** is acted upon to displace with respect to the surface of the annular base **11**. A distance of the assembling position of the piece **132** and the pull rod **131** from the surface of the annular base **11** will vary. Specifically, a distance of the piece **132** from the surface of the annular base **11** in the first state **116** is greater than the distance of the piece **132** from the surface of the annular base **11** in the second state **117**. Thus, when the annular base **11** is in the first state **116**, the piece **132** is acted upon by the pull rod **131** to press against the first clamping arm **112**, and when the annular base **11** is in the second state **117**, the piece **132** allows the first clamping arm **112** to back to the original position. On the other hand, in order to limit the assembling position of the cam portion **133**, the protective cover **10** includes an arc-shaped receiving groove **15** arranged on the first clamping arm **112** and facing the piece **132**, and the arc-shaped receiving groove **15** is arranged corresponding to the cam portion **133** and is provided for the cam portion **133** to be arranged therein, thereby providing a preliminary limit of the cam portion **133**.

(11) In addition, referring to FIG. 2, FIG. 3, and FIG. 4, in one embodiment, the pull rod **131** includes a coupling section **136** assembled with the piece **132**, a connecting section **137** extending from the coupling section **136** and penetrating through the through hole **114**, and a threading section **138** extending from the connecting section **137** and fixed in the assembly hole **115**. Furthermore, the pull rod **131** is assembled with the piece **132**. An opening **139** penetrates through the pull rod **131** and forms on the coupling section **136** of the pull rod **131**, and the opening **139** faces to two holes **140** of the piece **132**. The tightening component **13** comprises an assembly member **141** penetrating into the two holes **140** and the opening **139**. The assembly member **141** assembles with the pull rod **131** and the piece **132** to prevent the pull rod **131** from being disengaged from the piece **132**. In addition, the pull rod **131** is assembled in the assembly hole **115**, and the pull rod **131** is formed with a first thread **142** at the threading section **138**. An inner wall of the assembly hole **115** is formed with a second thread **118** corresponding to the first thread **142**, so that the threading section **138** is arranged in the assembly hole **115** while the first thread **142** is assembled with the second thread **118**.

(12) Accordingly, referring to FIG. 2, the tightening component **13** further comprises a gasket **143** arranged on the pull rod **131**, the gasket **143** is formed with an aperture **144** through which the pull rod **131** penetrates. The gasket **143** is provided between the piece **132** and the first clamping arm **112** and corresponds to the through hole **114**, and the gasket **143** contacts the cam portion **133** after assembly is completed.

(13) On the other hand, in one embodiment, the annular base **11** includes an abrasive belt supporting seat **119** extending from the second clamping arm **113** towards a direction opposite to the annular base **11**, the abrasive belt supporting seat **119** is assembled with an abrasive belt mounting component **40** of the abrasive belt hand tool **100**, so that the abrasive belt **30** is assembled with the abrasive belt mounting component **40** to be arranged on the protective cover **10**. In another embodiment, the annular base **11** includes a retaining wall **110** located on one side

opposite to the abrasive belt supporting seat **119**, and the retaining wall **110** prevents debris carried by the abrasive belt **30** from being ejected toward the user during operation.

Claims

1. A protective cover assembled on an abrasive belt hand tool, the protective cover assembled with a driving component of the abrasive belt hand tool and provided for an abrasive belt to move therein, the driving component provided with a driving shaft and a protective cover mounting portion formed at one end of the driving shaft, the protective cover comprising: an annular base, arranged on the protective cover mounting portion and provided for the driving shaft to extend into the annular base, the annular base comprising a first clamping arm and a second clamping arm facing the first clamping arm, a notch formed between the first clamping arm and the second clamping arm, the first clamping arm formed with a through hole, the second clamping arm formed with an assembly hole on a same extension line as the through hole, the assembly hole not penetrating through the second clamping arm, wherein a length of the assembly hole is longer than a length of the through hole, and an inner wall of the assembly hole is formed with a thread for assembling a threading section; and a tightening component, comprising a pull rod and a piece assembled with the pull rod, wherein the pull rod comprises a first end penetrating through the through hole and fixed in the assembly hole to pass through the notch, and a second end exposed on a surface of the first clamping arm, and wherein the piece is arranged on the surface of the first clamping arm, and wherein the pull rod is divided into a coupling section configured to be assembled with the piece, a connecting section extending from the coupling section and penetrating through the through hole, and the threading section extending from the connecting section and fixed in the assembly hole, and wherein the piece comprises a cam portion assembled with the second end of the pull rod and a shifting portion extended from the cam portion, and wherein an assembling position of the cam portion of the piece and the second end of the pull rod is deviated from a center of the cam portion, and the piece is configured to rotate with respect to a surface of the annular base by taking the assembling position as an axial center; wherein the annular base has a first state for tightening the protective cover mounting portion when the piece is forced to rotated toward the surface of the first clamping arm, in the first state, the second clamping arm is pulled by the first end of the pull rod and moved towards the first clamping arm to narrow the notch, the protective cover is fixed to the protective cover mounting portion in the first state; and wherein the annular base has a second state for releasing the protective cover mounting portion when the piece is forced to rotated opposite to the surface of the first clamping arm, in the second state, the second clamping arm is moved opposite to the first clamping arm to broaden the notch, the protective cover is disengaged with respect to the protective cover mounting portion in the second state.
 2. The protective cover assembled on an abrasive belt hand tool as claimed in claim 1, wherein the annular base comprises an abrasive belt supporting seat extending from the second clamping arm towards a direction opposite to the annular base.
 3. The protective cover assembled on an abrasive belt hand tool as claimed in claim 1, wherein the protective cover comprises an arc-shaped receiving groove arranged on the first clamping arm and facing the piece.
 4. The protective cover assembled on an abrasive belt hand tool as claimed in claim 1, wherein the annular base comprises an abrasive belt supporting seat extending from the second clamping arm towards a direction opposite to the annular base.
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